

Mega Assignment

A semester build on a topic you propose, and the three days it is graded on

Dates

Concept document · October 13 Pitch day · October 27 Poster session · December 3

Today

Twenty five minutes on what the thing is, then you pair up and start arguing about topics.

Christoffer Heckman

Department of Computer Science, University of Colorado Boulder

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Quick overview

What it is. One artifact, built over nine weeks, on a question you choose.

Twenty points, three deliverables

The concept document. Four questions, two pages, due October 13.

The compute question is the one people skip

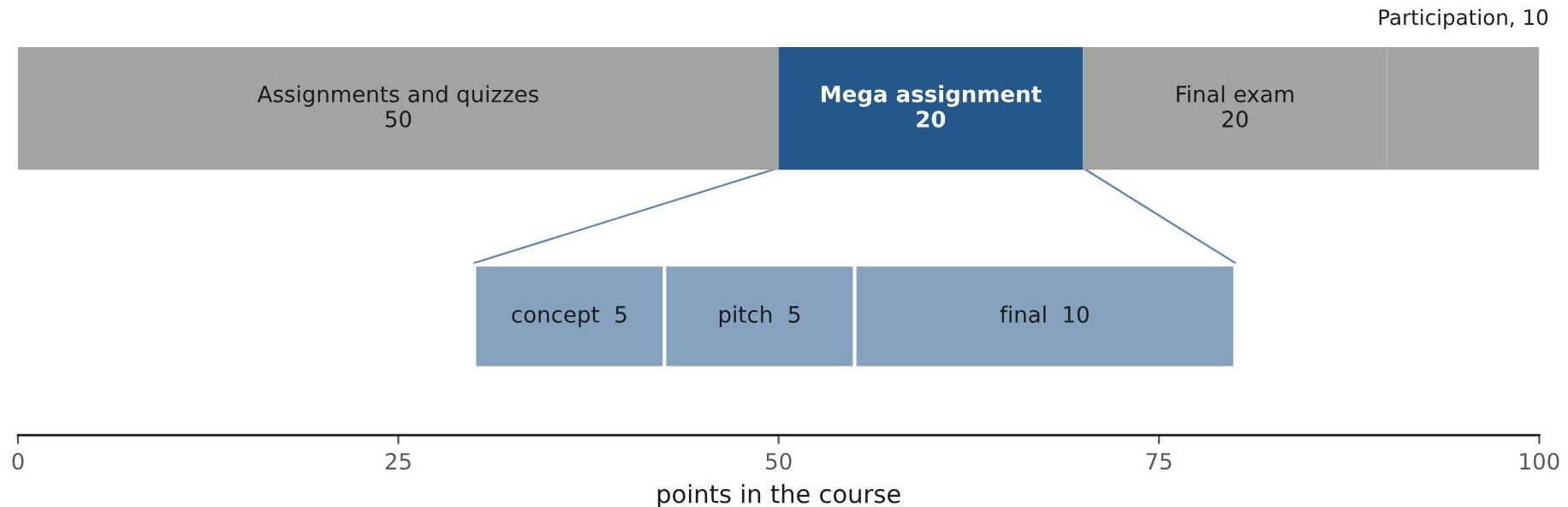
Scoping. How big a project should be.

Worked against real directions

Teams and topics. Pairs, and eight directions out of the assignments.

Then the rest of the hour to form them

Twenty Points



Concept document, 5

Two pages, October 13. Graded on whether the plan can be answered. An ambitious plan with no measurement in it scores below a modest one.

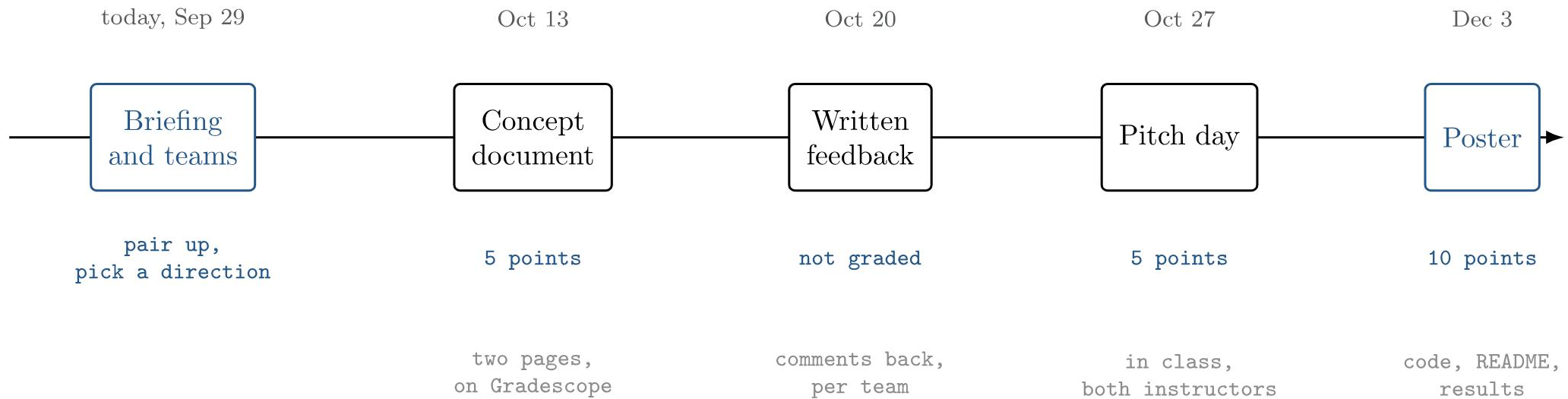
Pitch day, 5

In class, October 27. You present the plan and defend it in front of both of us.

Final submission, 10

Code, README and results, at the poster session on December 3.

The Calendar



the Oct 20 comments are the one ungraded
stop: a plan changed here costs nothing,
a plan changed in November costs build time

1

THE CONCEPT DOCUMENT

Four Questions

Four Questions

SECTION 1

What you will build

A router, a quantized checkpoint, an eval harness, a dataset.

weak answer:
“study X”

SECTION 2

On what data

Which corpus, how big, where it comes from, and what you do when it is not enough.

weak answer:
“a public dataset”

SECTION 3

On what compute

Which machines, how many GPU-hours, and the arithmetic that got you that number.

weak answer:
“the course cluster”

SECTION 4

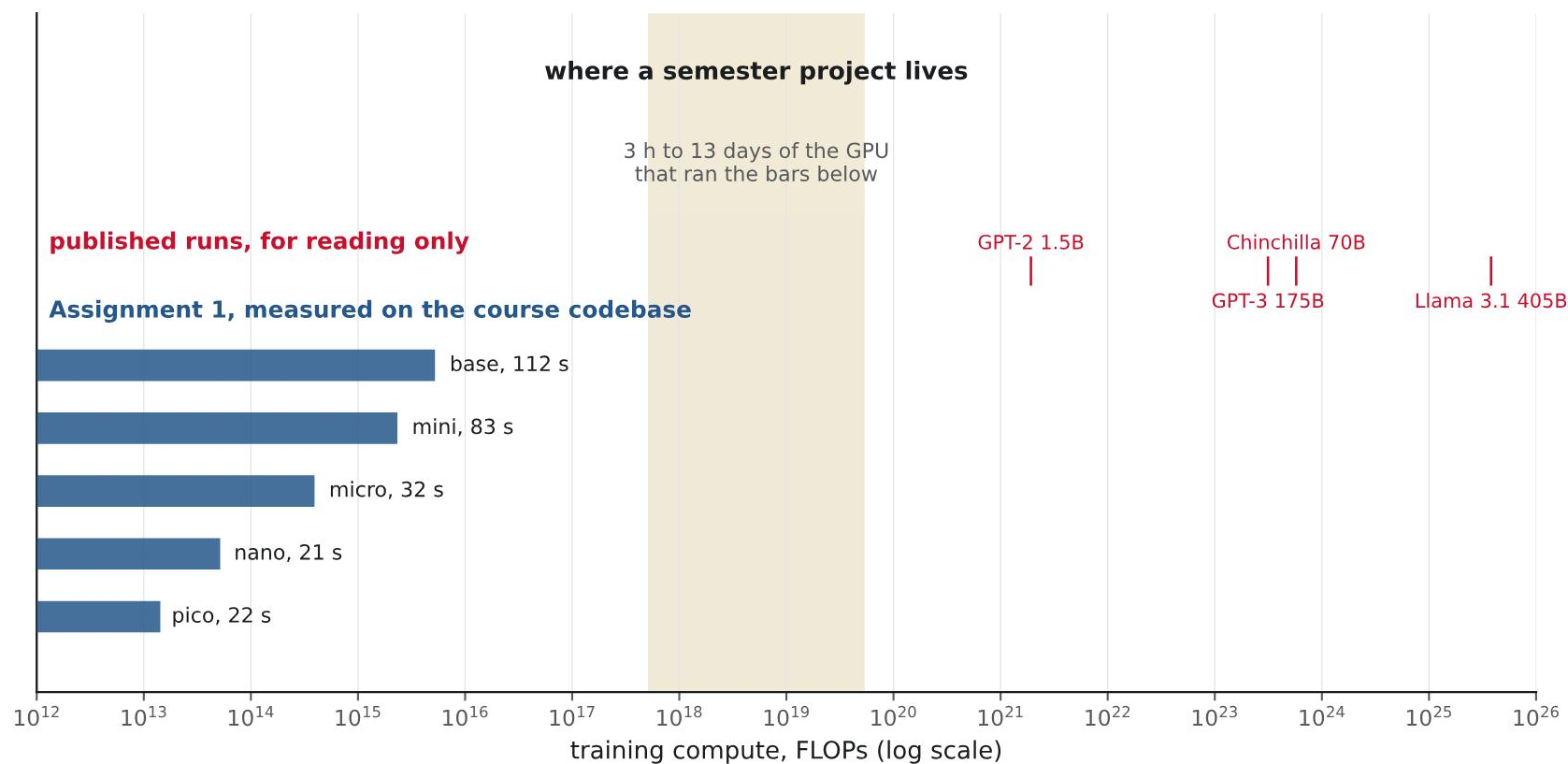
How you will know it worked

Which measurement, which baseline, and the number that would make you abandon the idea.

weak answer:
“evaluate performance”

two pages, due October 13, 5 points
graded on whether section 4 names the number that would make you stop

Compute Arithmetic



What you have: Google Cloud credits, Chameleon, NSF ACCESS on Delta, and the course cluster, four RTX 8000 at 48 GB. Shared with everyone in the room.

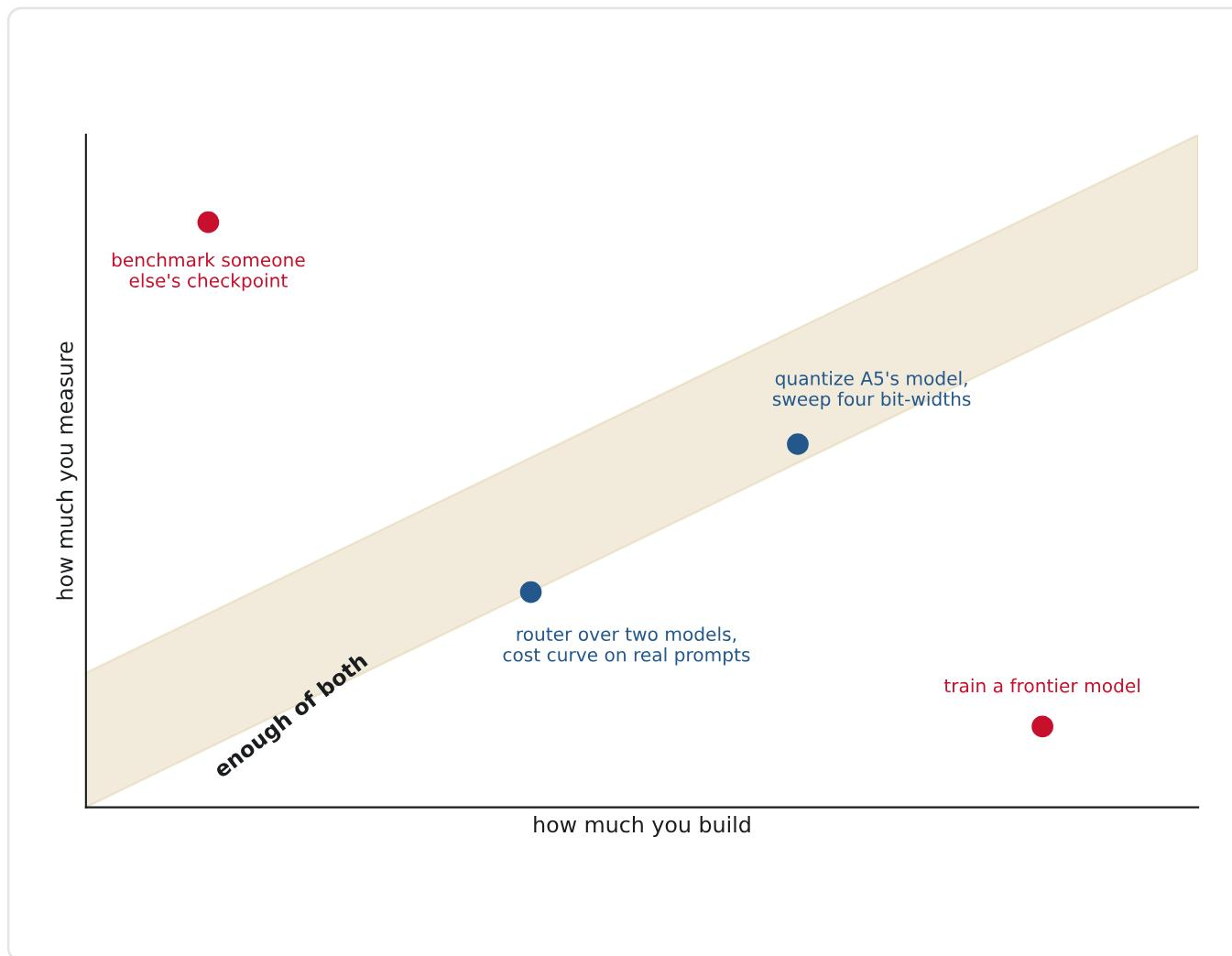
What section three needs: a GPU-hour number and the arithmetic that produced it, from model size, tokens, and a throughput you measured yourself.

2

SCOPING AND TOPICS

The Right Size

Too Big, Too Small

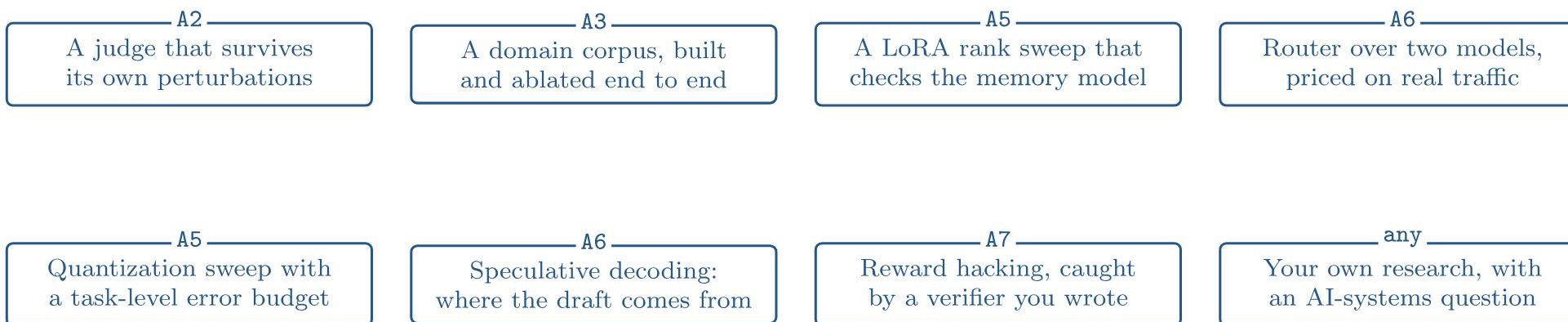
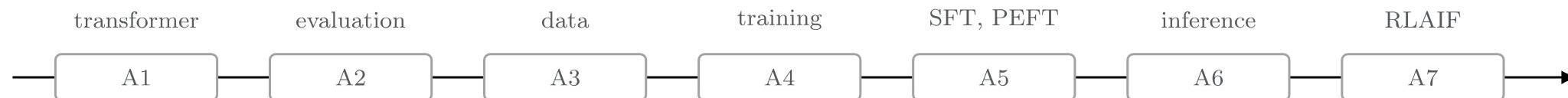


Too big. Anything whose first sentence is "train a model that". You have five weeks of building and a shared cluster.

Too small. Anything you can finish by running someone else's evaluation script on someone else's checkpoint.

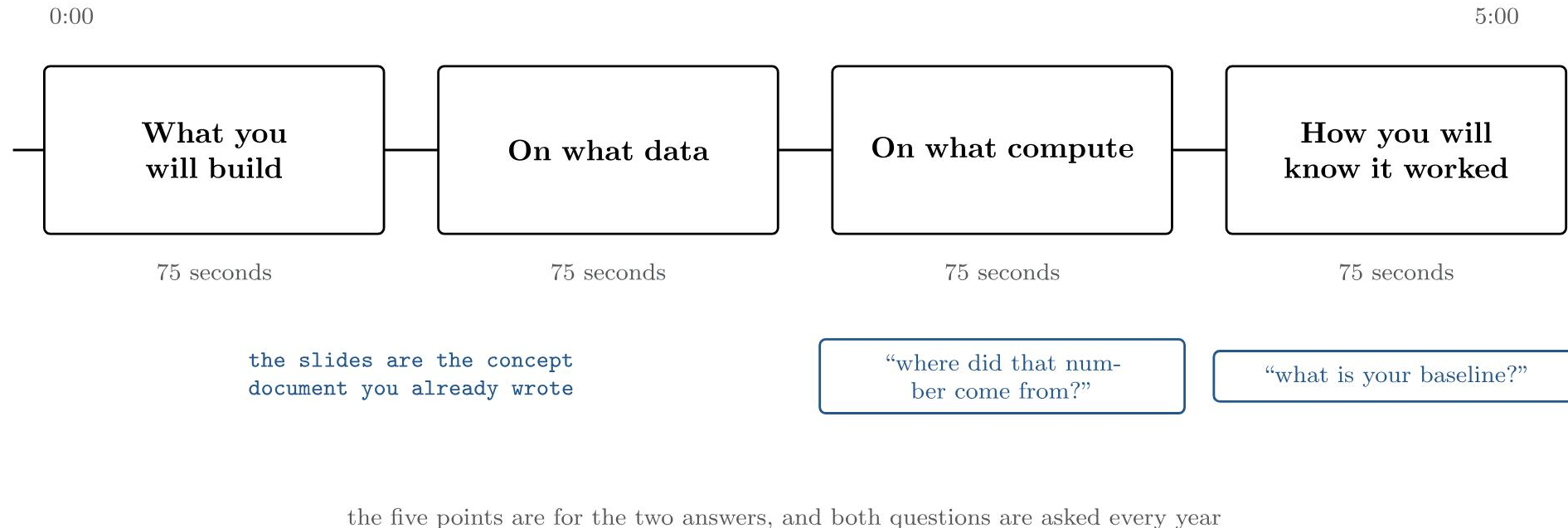
The band. You build one thing and measure it against something, so the project has a position on both axes.

Eight Directions



each of these is a direction; the concept document is where one becomes a project
the last one is for anyone already doing research: bring it, and point it at an AI-systems question

Pitch Day



Five minutes

Four slides, both instructors in the room. We run it tight so every pair goes.

We interrupt

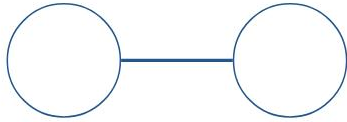
That is the assessment. The five points are the defense.

No new deck

The four slides are the four sections you already wrote.

Teams

The default



two people, both writing code
nothing to arrange

The exception



faculty advisor

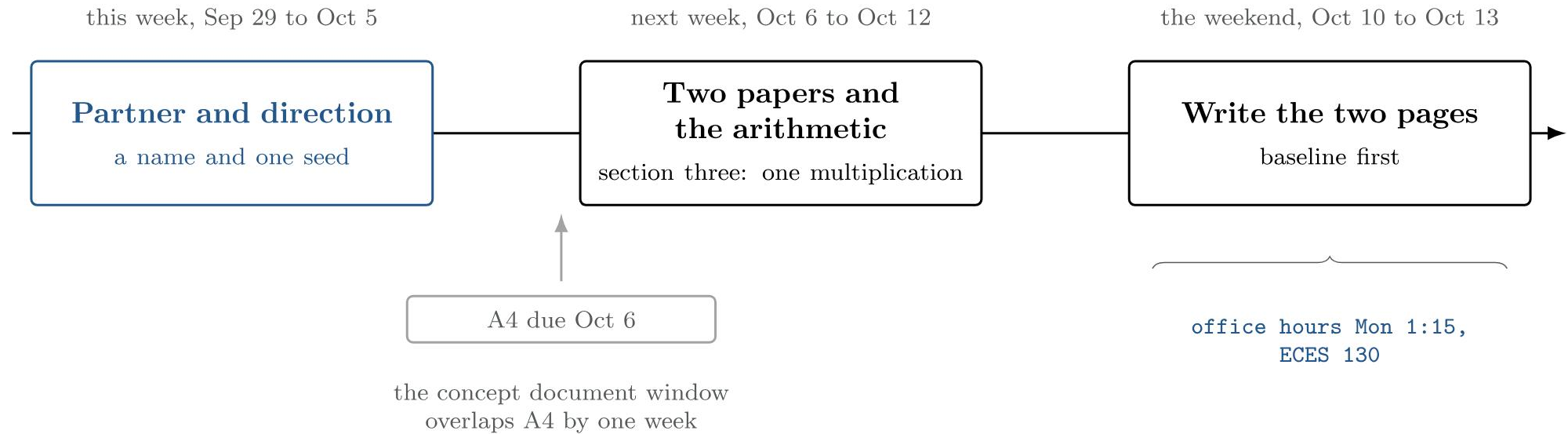
alone, when the project is research
you are already doing

teams form today

teams freeze Oct 13

the concept document is where you declare which one you are

Before October 13



before you leave today, name the pair and the direction, even if both change by the 13th

Now

Find a partner. Pick a seed. Write down the baseline you would measure it against.

I will come around. If you get to a baseline you believe in, you have already written a third of the concept document.